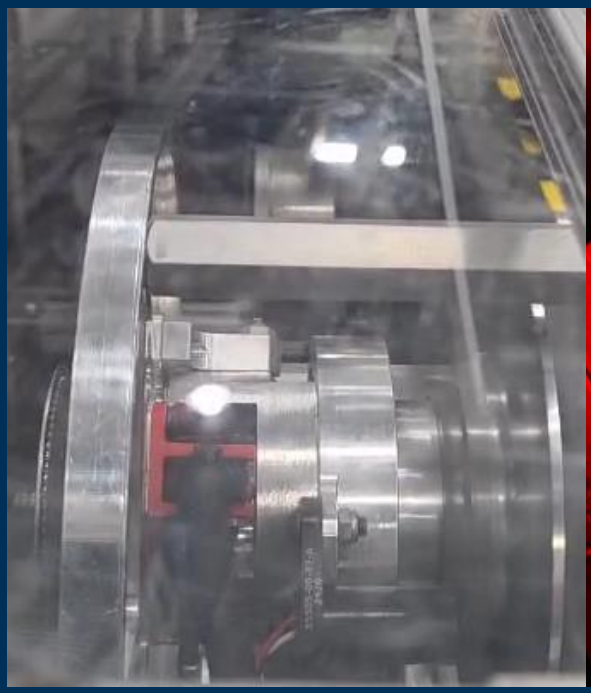
Naitik Mundra

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B.S. Mechanical Engineering + B.S. Physics
Georgia Institute of Technology, GPA 4.0

Mechanical Engineering and Mechatronics Portfolio





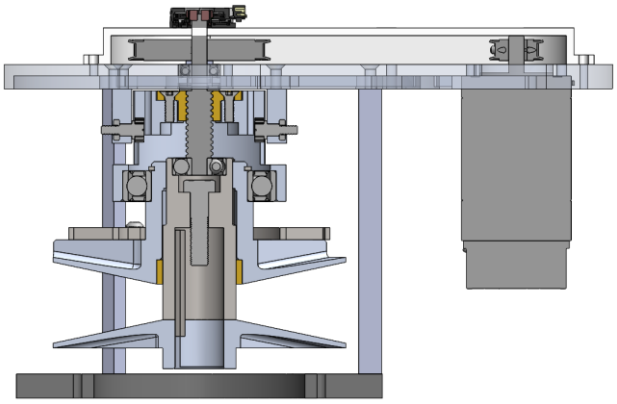
I **design in CAD**, **verify in FEA**, then prototype and **machine it myself**. I **wire and calibrate the sensors** and **tune the controls**.

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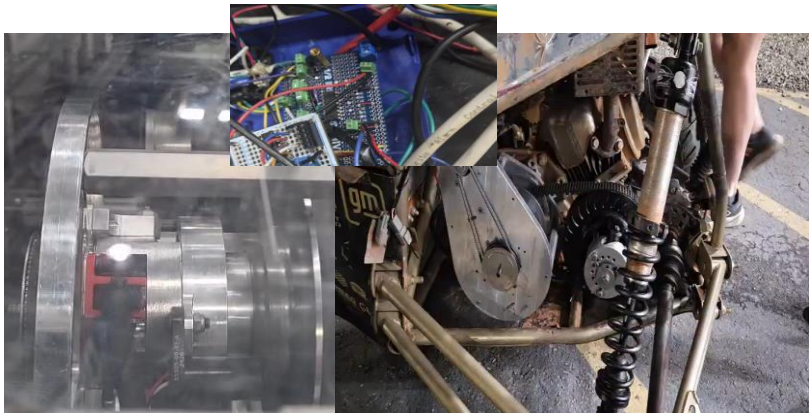
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eCVT: ELECTRONIC CONTINUOUSLY VARIABLE TRANSMISSION



eCVT assembly, SolidWorks (*Not my CAD)



eCVT mounted on dyno (left) and car (right)

What?

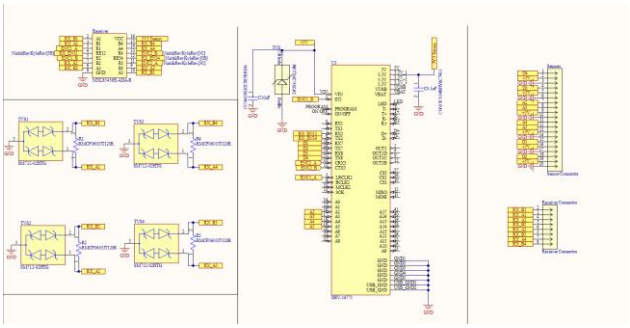
- Electronic CVT for the Baja SAE car that shifts 2x faster, led as Project Lead
- Teensy 4.1 control, tested from gain-only through PID strategies.

How?

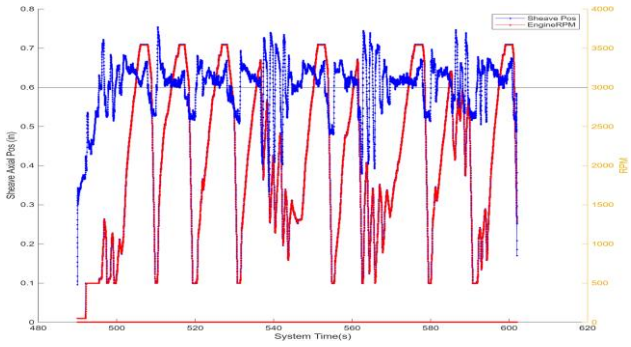
- Built the dyno fixture and safety shielding, then wired the prototype on perf-board.
- Integrating 4 sensors, reading at over 20K/s
- 4-layer Altium PCBs

Results

- 2x faster shifting, verified on the dyno.
- Rotational Inertia reduced 62% over mechanical CVT
- EMI in the readings cut by over 80%.



Altium PCB Schematic



Analysis of testing runs in MATLAB

Other work:

- Debugged EMI and introduced shielded wire
- Presented PDR and CDR
- Wrote 5+ test plans

Control strategies tried:

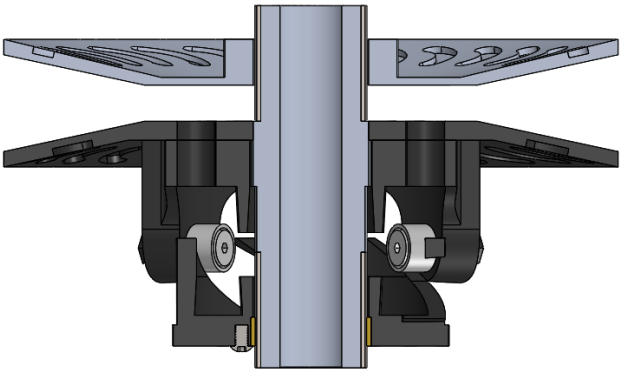
- Basic gain control
- PID on the Teensy 4.1

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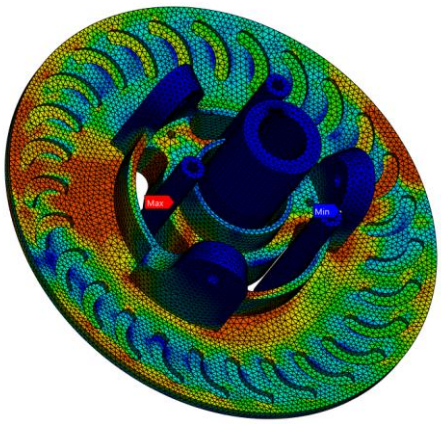
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CUSTOM SECONDARY DRIVEN PULLEY (~10% LESS MASS)



CAD of CVT Driven (Secondary) Pulley



Optimized in Ansys

What?

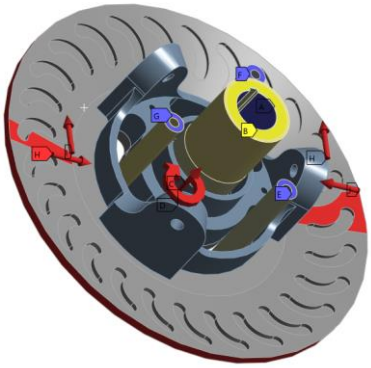
- Lighter driven pulley for the eCVT
- Cutouts in Sheave Face remove mass

How?

- Teardown of COTS part; Root Cause Analysis
- Hand Calcs (Shigley's)
- Ansys FEA

Results

- 10% Less Mass
- 12% Less Rotational Inertia



Load Cases Derived From Shigley

Other work:

- Analysis of competition footage to understand back shifting and tuning concerns



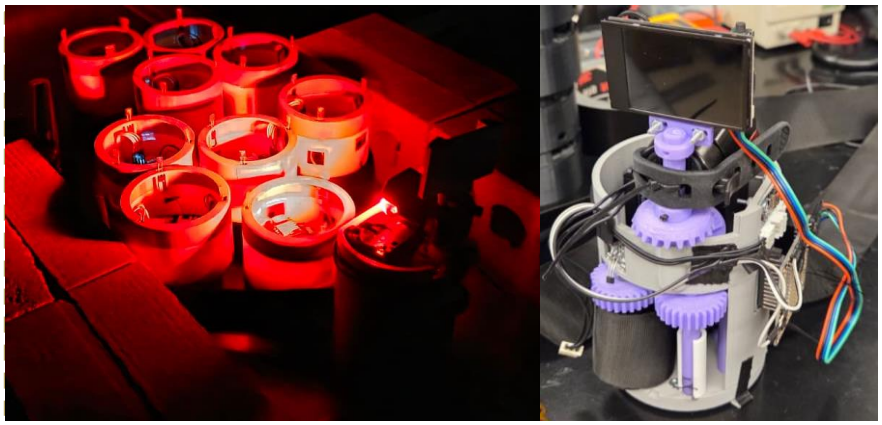
Gaged CVT that was modified after teardown

NAITIK MUNDRA

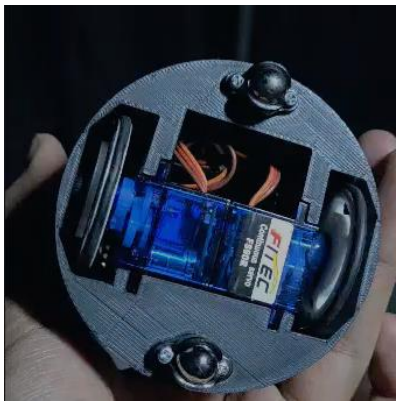
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SWARM ROBOTS: ACTIVE MATTER RESEARCH (CRAB LAB)



Left:Swarm of 10 Robots during Run; Right: Belt Station



Slip clutch protecting servos

What?

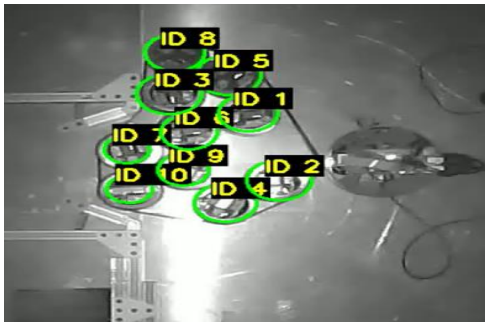
- 10 swarm robots built in one month for active matter trials: soldered, machined and programmed each.
- RL for physical AI, optimal control of compliant multi-agent robots.

How?

- Designed a 2 mm slip clutch to protect 20 DC servo motors.
- Measured belt tension from calibrated motor PID torque instead of strain gauges.
- Designed 6 gear belt spooling mechanism

Results

- Stall incidents down 85%.
- Belt spooling: 90% less slip, 35% less mass.
- 50+ pose-tracked trials (40+ video hours), >95% accuracy;
- 250+ simulator runs.



Computer Vision Tracking



Door test rig for swarm manipulation

Tools built:

- Python/OpenCV pose tracking, calibrated to physical units
- C++/MATLAB simulator with implicit integration

Test rigs:

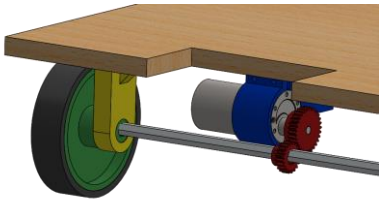
- Spring-loaded doors that open under adjustable finite push and close on release
- Belt tension measurement from motor torque

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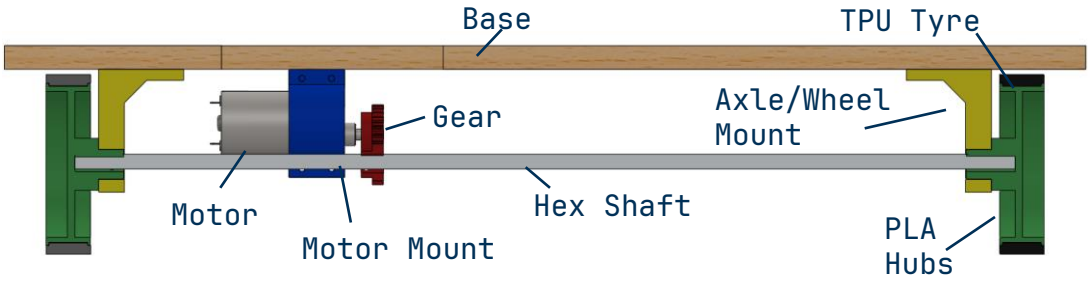
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ME2110 DRIVETRAIN FOR AUTONOMOUS ROBOT



Isometric View: Assembly



Section View: Assembly

What?

- Drivetrain for the ME2110 Competition
- Sized From Torque And Rolling Resistance Calculations

How?

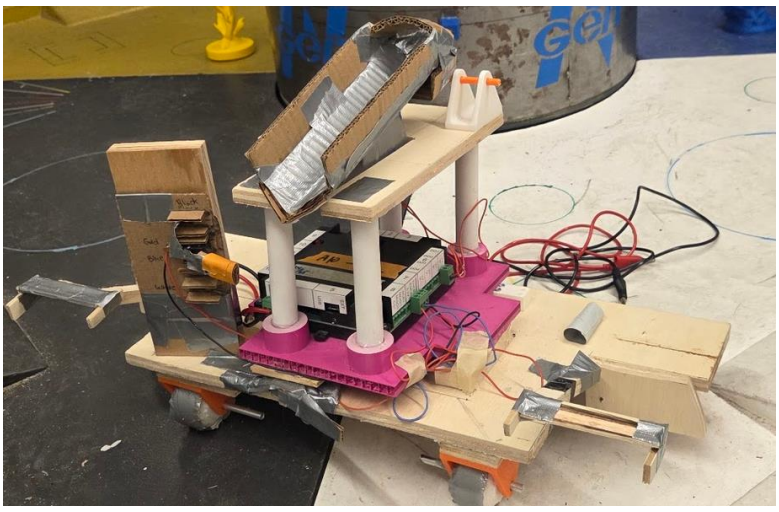
- Gear Sizing using torque hand-calcs
- Set Screw locking gears to axle
- Fully 3D printable wheels, hubs, and mounts

Results

- Fastest Robot
- 100% Success Rate
- Weight < 150g
- Assembly Time < 20 minutes

	A	B	C	D	E	F
1	**Constants**				**Required Force**	
2	mass	0.8000	kg		safety factor	1.2500
3	g	9.8100	m/s^2		F_forward = SF * C_rr * m * g	0.7848
4	wheel radius	0.0445	m			
5	rolling resistance coeff	0.0800				
6	efficiency	0.5000				
7	motor rated torque	0.1275	(N*m)			
8						
9						
10	**Drivetrain Calculations**					
11	Gear Ratio	1.5				
12	Required Torque = F_forward * whee radius * gear ratio/efficiency	0.1048				

Excel Hand Calcs



Prototype Robot Built in <18 Hours

Documentation:

- HOQ: 10 customer requirements to 60 engineering requirements
- Engineering drawings

Let's build something.

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